

Linear Statistical Models

Week 11 Graded Assignment Solution

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December 7, 2025

Question 1

Given Model:

$$\underset{\sim}{Y} = \underset{\sim}{X}\underset{\sim}{\beta} + \underset{\sim}{\epsilon}$$

where

$$\underset{\sim}{Y} = \begin{pmatrix} 80 \\ 70 \\ 60 \end{pmatrix}, \quad \underset{\sim}{X} = \begin{pmatrix} 1 & 6 \\ 1 & 6 \\ 1 & 5 \end{pmatrix}$$

and ϵ 's are uncorrelated random variables with variance σ^2 .

(a) Least Square Estimate for β

The least squares estimator is given by the normal equation:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Step 1: Compute $X^T X$

$$X^T X = \begin{pmatrix} 1 & 1 & 1 \\ 6 & 6 & 5 \end{pmatrix} \begin{pmatrix} 1 & 6 \\ 1 & 6 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 17 \\ 17 & 97 \end{pmatrix}$$

Step 2: Compute $(X^T X)^{-1}$ Determinant $|X^T X| = (3)(97) - (17)(17) = 291 - 289 = 2$.

$$(X^T X)^{-1} = \frac{1}{2} \begin{pmatrix} 97 & -17 \\ -17 & 3 \end{pmatrix}$$

Step 3: Compute $X^T Y$

$$X^T Y = \begin{pmatrix} 1 & 1 & 1 \\ 6 & 6 & 5 \end{pmatrix} \begin{pmatrix} 80 \\ 70 \\ 60 \end{pmatrix} = \begin{pmatrix} 210 \\ 1200 \end{pmatrix}$$

Step 4: Solve for $\hat{\beta}$

$$\hat{\beta} = \frac{1}{2} \begin{pmatrix} 97 & -17 \\ -17 & 3 \end{pmatrix} \begin{pmatrix} 210 \\ 1200 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} (97)(210) + (-17)(1200) \\ (-17)(210) + (3)(1200) \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 20370 - 20400 \\ -3570 + 3600 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -30 \\ 30 \end{pmatrix} = \begin{pmatrix} -15 \\ 15 \end{pmatrix}$$

$$\boxed{\hat{\beta} = \begin{pmatrix} -15 \\ 15 \end{pmatrix}}$$

(b) Fitted Regression Line and Prediction

The fitted model is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$.

$$\hat{y} = -15 + 15x$$

Prediction for $x = 5.5$:

$$\hat{y}_{new} = -15 + 15(5.5) = -15 + 82.5 = \mathbf{67.5}$$

(c) Prediction Error (Residuals)

Compute $\hat{y}_i$ for each observation:

- Obs 1: $x = 6, y = 80 \Rightarrow \hat{y} = -15 + 15(6) = 75$. Error $e_1 = 80 - 75 = 5$.
- Obs 2: $x = 6, y = 70 \Rightarrow \hat{y} = -15 + 15(6) = 75$. Error $e_2 = 70 - 75 = -5$.
- Obs 3: $x = 5, y = 60 \Rightarrow \hat{y} = -15 + 15(5) = 60$. Error $e_3 = 60 - 60 = 0$.

Prediction errors are: **5, -5, 0**.

(d) Compute RSS and TSS

RSS (Residual Sum of Squares): $\sum e_i^2$

$$RSS = (5)^2 + (-5)^2 + 0^2 = 25 + 25 = \mathbf{50}$$

TSS (Total Sum of Squares): $\sum (y_i - \bar{y})^2$

$$\bar{y} = \frac{80 + 70 + 60}{3} = 70$$

$$TSS = (80 - 70)^2 + (70 - 70)^2 + (60 - 70)^2 = 100 + 0 + 100 = \mathbf{200}$$

(e) Estimate of σ^2

$$\hat{\sigma}^2 = \frac{RSS}{n - p}$$

Here $n = 3$ and $p = 2$ (intercept + slope).

$$\hat{\sigma}^2 = \frac{50}{3 - 2} = \mathbf{50}$$

(f) Coefficient of Determination R^2

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{50}{200} = 1 - 0.25 = \mathbf{0.75}$$

Interpretation: 75% of the total variation in Y is explained by the linear relationship with X .

(g) Find $Var(\hat{\beta}_1)$

The variance-covariance matrix of $\hat{\beta}$ is:

$$Var(\hat{\beta}) = \hat{\sigma}^2(X^T X)^{-1}$$

$$Var(\hat{\beta}) = 50 \times \begin{pmatrix} 48.5 & -8.5 \\ -8.5 & 1.5 \end{pmatrix}$$

The variance of the slope $\hat{\beta}_1$ corresponds to the element at (2, 2):

$$Var(\hat{\beta}_1) = 50 \times 1.5 = \mathbf{75}$$

(h) Hypothesis Testing

Test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$ at $\alpha = 0.05$.

(i) t-statistic:

$$SE(\hat{\beta}_1) = \sqrt{Var(\hat{\beta}_1)} = \sqrt{75} \approx 8.66$$

$$t_{stat} = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} = \frac{15}{8.66} \approx \mathbf{1.732}$$

(ii) Conclusion using R criteria:

```
t_calculated <- 1.732 # The t-statistic we found manually
n <- 3 # Sample size
p <- 2 # Number of parameters (beta0, beta1)
alpha <- 0.05 # Significance level (5%)

df <- n - p
print(paste("Degrees of Freedom:", df))

# For a two-tailed test, we look at 1 - alpha/2 (i.e., 0.975)
t_critical <- qt(1 - alpha / 2, df)

cat("\n--- Hypothesis Test Results ---\n")
cat("Calculated t-statistic: ", t_calculated, "\n")
cat("Critical t-value:      ", t_critical, "\n")

if (abs(t_calculated) > t_critical) {
  print("Conclusion: REJECT Null Hypothesis (Significant
    Relationship)")
} else {
```

```
print("Conclusion: FAIL TO REJECT Null Hypothesis (No
      Significant Relationship)")
}
```

```
r> t_calculated <- 1.732 # The t-statistic we found manually
n <- 3 # Sample size
p <- 2 # Number of parameters (beta0, beta1)
alpha <- 0.05 # Significance level (5%)

df <- n - p
print(paste("Degrees of Freedom:", df))

# For a two-tailed test, we look at 1 - alpha/2 (i.e., 0.975)
t_critical <- qt(1 - alpha / 2, df)

cat("\n--- Hypothesis Test Results ---\n")
cat("Calculated t-statistic: ", t_calculated, "\n")
cat("Critical t-value:      ", t_critical, "\n")

if (abs(t_calculated) > t_critical) {
  print("Conclusion: REJECT Null Hypothesis (Significant Relationship)")
} else {
  print("Conclusion: FAIL TO REJECT Null Hypothesis (No Significant Relationship)")
}

[1] "Degrees of Freedom: 1"
--- Hypothesis Test Results ---
Calculated t-statistic: 1.732
Critical t-value:      12.7062
[1] "Conclusion: FAIL TO REJECT Null Hypothesis (No Significant Relationship)"
```

Degrees of freedom $df = n - p = 3 - 2 = 1$. The critical t-value for a two-tailed test at $\alpha = 0.05$ with $df = 1$ is approximately **12.706**.

$$|t_{stat}| = 1.732 < 12.706$$

Since the calculated t-statistic is less than the critical value, we **fail to reject** H_0 .
Conclusion: There is no significant linear relationship between X and Y .

(i) R Code Implementation

```
# Define data
Y <- c(80, 70, 60)
X <- c(6, 6, 5)

# Fit linear model
model <- lm(Y ~ X)

# Summary to check hypothesis
summary(model)
```

Comment: The `summary(model)` output in R will show a p-value for the slope coefficient. Given our manual calculation, the p-value will be greater than 0.05 (specifically approx 0.33), confirming that we fail to reject H_0 . Hence no, significant Linear Relationship between X and Y

```
r> Y <- c(80, 70, 60)
X <- c(6, 6, 5)
model <- lm(Y ~ X)

# Summary to check hypothesis
summary(model)

Call:
lm(formula = Y ~ X)

Residuals:
    1     2     3 
5.00e+00 -5.00e+00  5.16e-15 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -15.00      49.24  -0.305   0.812
X              15.00       8.66   1.732   0.333

Residual standard error: 7.071 on 1 degrees of freedom
Multiple R-squared:  0.75,    Adjusted R-squared:  0.5 
F-statistic: 3 on 1 and 1 DF,  p-value: 0.3333
```

Question 2: Battery Life Analysis

1. Hypotheses

We are testing the variance components (Random Effects Model):

- $H_0 : \sigma_\tau^2 = 0$ (The variance component among battery brands is zero).
- $H_1 : \sigma_\tau^2 > 0$ (The variance component among battery brands is greater than zero).

2. Analysis Method

Since the brands are chosen randomly from a population of brands, this is a **Random Effects One-Way ANOVA**. In a balanced One-Way ANOVA, the test statistic $F = MS_{Treatments}/MS_{Error}$ is used to test H_0 .

3. R Code for Solution

Since the dataset `Battery.txt` is external, the following R code is required to perform the analysis and obtain the numerical results:

```
# w10Q2

data <- read.table("21F1001107_10/data/Battery.txt", header =
  TRUE)
View(data)

data$trt <- as.factor(data$trt)

# For a balanced design, aov() is sufficient for the F-test
model <- aov(lifetime ~ trt, data = data)

summary(model)

anova_table <- summary(model)[[1]]
MS_Brand <- anova_table["trt", "Mean Sq"]
MS_Error <- anova_table["Residuals", "Mean Sq"]
n <- 4 # Number of replications

sigma_brand_sq <- (MS_Brand - MS_Error) / n
cat("Estimated Variance Component for Brand:", sigma_brand_sq)
```

(row)	trt	lifetime
1	BrandA	110
2	BrandA	113
3	BrandA	108
4	BrandA	115
5	BrandB	118
6	BrandB	116
7	BrandB	112
8	BrandB	117
9	BrandC	108
10	BrandC	107
11	BrandC	112
12	BrandC	108
13	BrandD	117
14	BrandD	112
15	BrandD	115
16	BrandD	119

```

r$> # 1. Import Data
data <- read.table("21F1001107_10/data/Battery.txt", header = TRUE)
View(data)
# 2. Ensure 'Brand' is treated as a factor
data$trt <- as.factor(data$trt)

# 3. Perform ANOVA
# For a balanced design, aov() is sufficient for the F-test
model <- aov(lifetime ~ trt, data = data)

# 4. View Results
summary(model)

anova_table <- summary(model)[[1]]
MS_Brand <- anova_table["trt", "Mean Sq"]
MS_Error <- anova_table["Residuals", "Mean Sq"]
n <- 4 # Number of replications

sigma_brand_sq <- (MS_Brand - MS_Error) / n
cat("Estimated Variance Component for Brand:", sigma_brand_sq)

      Df Sum Sq Mean Sq F value    Pr(>F)
trt      3 141.69   47.23    6.211 0.00863 **
Residuals 12  91.25    7.60
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Estimated Variance Component for Brand: 9.90625

```

4. Conclusion Criteria

- If the **p-value** in the summary output is less than $\alpha = 0.05$, we **Reject** H_0 and conclude there is significant variation between battery brands.

Since the p-value is 0.00863 which is less than 0.05 therefore we **Reject** H_0 and conclude **Not all brands have the same average lifetime. At least one brand differs significantly.**

Question 3: Loom Fabric Strength

Problem Statement: Four looms were chosen randomly, and four observations of fabric strength were made on each. We test if the variance component among looms is significant.

1. Data Summary

Loom	Obs 1	Obs 2	Obs 3	Obs 4	Total (T_i)
1	98	97	99	96	390
2	91	90	93	92	366
3	96	95	97	95	383
4	95	96	99	98	388
Grand Total (G)					1527

Number of groups (a) = 4, Replicates (n) = 4, Total observations (N) = 16.

2. Hypotheses

- $H_0 : \sigma_{\text{loom}}^2 = 0$ (Variance component among looms is zero).
- $H_1 : \sigma_{\text{loom}}^2 > 0$ (Variance component among looms is positive).

3. ANOVA Calculations

Step 1: Correction Factor (CF)

$$CF = \frac{G^2}{N} = \frac{1527^2}{16} = \frac{2331729}{16} = 145733.0625$$

Step 2: Total Sum of Squares (SS_T)

$$\sum y_{ij}^2 = 98^2 + 97^2 + \dots + 98^2 = 145845$$

$$SS_T = 145845 - 145733.0625 = \mathbf{111.9375}$$

Step 3: Treatment (Between Looms) Sum of Squares (SS_{Tr})

$$\begin{aligned} SS_{Tr} &= \frac{\sum T_i^2}{n} - CF = \frac{390^2 + 366^2 + 383^2 + 388^2}{4} - CF \\ &= \frac{152100 + 133956 + 146689 + 150544}{4} - 145733.0625 \\ &= \frac{583289}{4} - 145733.0625 = 145822.25 - 145733.0625 = \mathbf{89.1875} \end{aligned}$$

Step 4: Error Sum of Squares (SS_E)

$$SS_E = SS_T - SS_{Tr} = 111.9375 - 89.1875 = \mathbf{22.75}$$

4. ANOVA Table

Source	DF	SS	MS	F-ratio	P-value
Between Looms	3	89.1875	29.729	15.68	< 0.001
Error (Within)	12	22.7500	1.896		
Total	15	111.9375			

5. Conclusion

The calculated F-value is **15.68**. Using a standard F-distribution table (or R), the critical value $F_{0.05,3,12}$ is approximately **3.49**.

Since $F_{calc} = 15.68 > 3.49$ (and the p-value is approx 0.000188), we **Reject the Null Hypothesis**.

Final Statement: There is significant evidence at the 5% level to conclude that the variance component among the looms is greater than zero ($\sigma_{loom}^2 > 0$). This means the looms differ significantly in their performance.